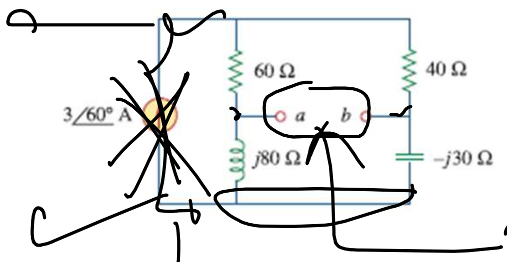
10.64 For the circuit shown in Fig. 10.107, find the Norton equivalent circuit at terminals *a*-*b*.



Soln:

Z_N, I_N

$Z_{eq} = Z_N$

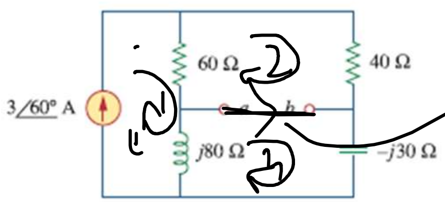
$$Z_N = (60 + 40) \parallel (j80 - j30)$$

$=$

$\underline{\underline{I_N}}$

By applying mesh analysis.

10.64 For the circuit shown in Fig. 10.107, find the Norton equivalent circuit at terminals *a*-*b*.



$\rightarrow I_{sc} = I_N.$

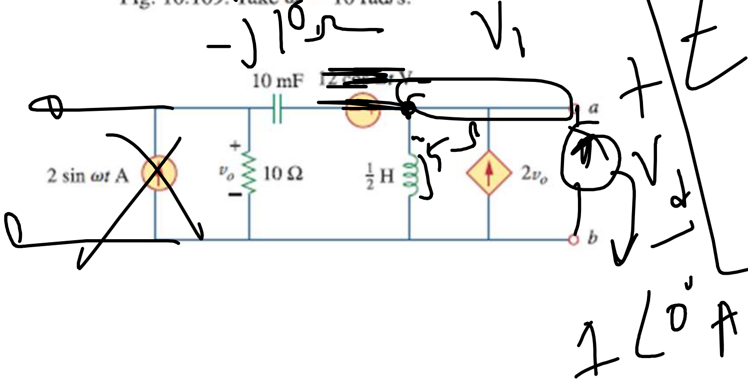
$I_1 = 3\angle 60^\circ$

$I_2 = ?$

$I_3 = \ldots$

$I_N = I_3 - I_2$

10.66 At terminals *a-b*, obtain Thevenin and Norton equivalent circuits for the network depicted in Fig. 10.109. Take $\omega = 10$ rad/s.



$$\boxed{Z_N = Z_{th}}$$

applying KCL at node of V_1

$$\frac{V_1}{10-j10} + \frac{V_1}{5j} - 2\times\left[\sqrt{2}\angle 45^\circ\right]V_1 - 1\angle 0^\circ = 0$$

$$V_0 = V_{10\Omega} = \frac{10}{10-j10} \times V_1$$

$$= \left(\sqrt{2}\angle 45^\circ\right) V_1$$

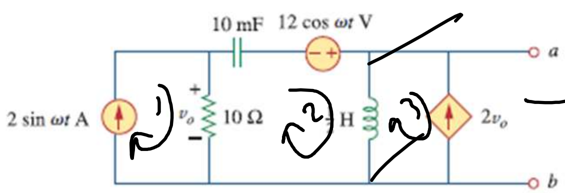
$$\therefore V_1 = \frac{1\angle 0^\circ}{\left(\frac{1}{10-j10} + \frac{1}{5j} - 2\sqrt{2}\angle 45^\circ\right)} = \ldots$$

$$V_1 = V_d$$

$$\therefore Z_N = \frac{V_d}{1\angle 0^\circ}$$

$$I_N = \frac{V_{th}}{Z_N}$$

10.66 At terminals *a-b*, obtain Thevenin and Norton equivalent circuits for the network depicted in Fig. 10.109. Take $\omega = 10$ rad/s.



$\to V_{oc}$

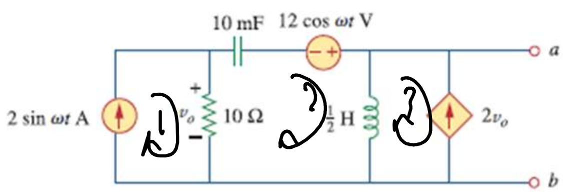
$$I_1 = 2\angle 0^\circ$$

$$I_3 = -2V_o = -20(I_1 - I_2)$$

$$= -20(2\angle 0^\circ - I_2)$$

$$V_o = (I_1 - I_2) \times 10$$

10.66 At terminals *a*-*b*, obtain Thevenin and Norton equivalent circuits for the network depicted in Fig. 10.109. Take $\omega = 10$ rad/s.



KVL at mesh-2

$$(10 - j10) \times (I_2 - I_1) - 12\angle 0^\circ + j5 \times (I_2 - I_3)$$

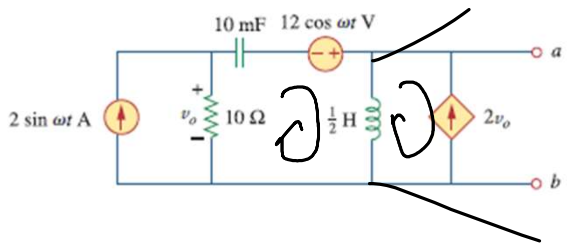
$$\Rightarrow (10 - j10) \times (I_2 - 2\angle 0^\circ) - 12\angle 0^\circ + j5 \times \left(I_2 + 20(2\angle 0^\circ - I_2)\right) = 0$$

$$I_2(10 - j10) - 2\angle 0^\circ(10 - j10) - 12\angle 0^\circ$$

$$+ j5 I_2 + 46\angle 0^\circ - 20I_2 = 0$$

$$\therefore \quad I_2 = \frac{21\angle 68^\circ}{30\angle -9.5^\circ} = 0.7\angle 80^\circ$$

10.66 At terminals *a-b*, obtain Thevenin and Norton equivalent circuits for the network depicted in Fig. 10.109. Take $\omega = 10$ rad/s.



$$I_3 = -20(2\angle 0^\circ - I_2)$$

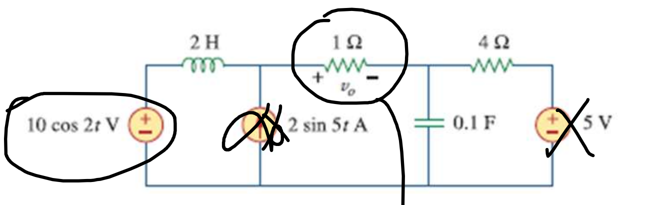
$$= 40\angle 160^\circ$$

$$V_{th} = (I_2 - I_3) \times j5 = 200\angle 71^\circ \text{ V}$$

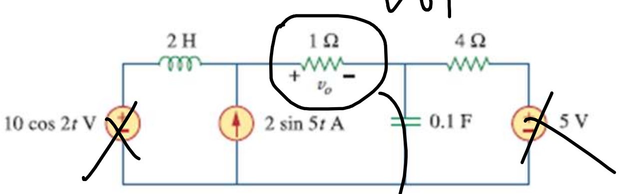
$$I_N = \frac{V_{th}}{Z_N}$$

Superposition theorem

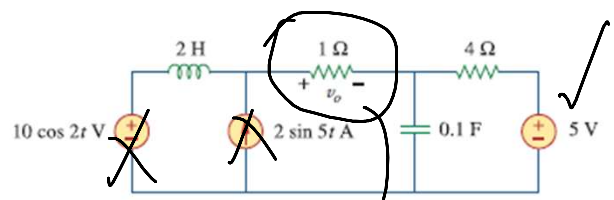
We turn off only the independent source.



V_{o1}



V_{o2}

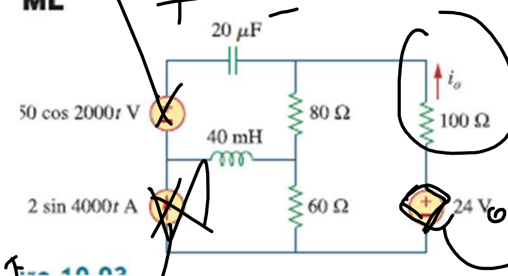


V_{o3}

$$V_o = V_{o1} + V_{o2} + V_{o3}$$

ON

+ V_o −



OFF

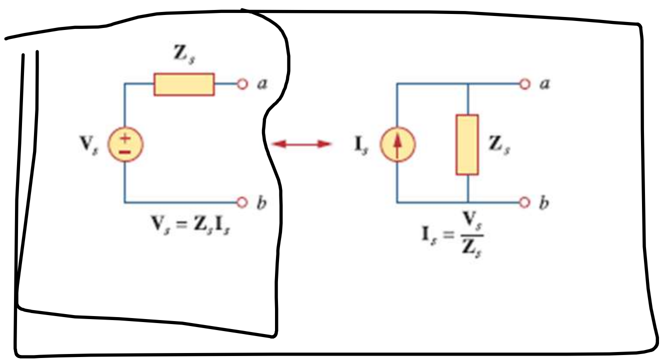
suppose it is a dependent voltage source

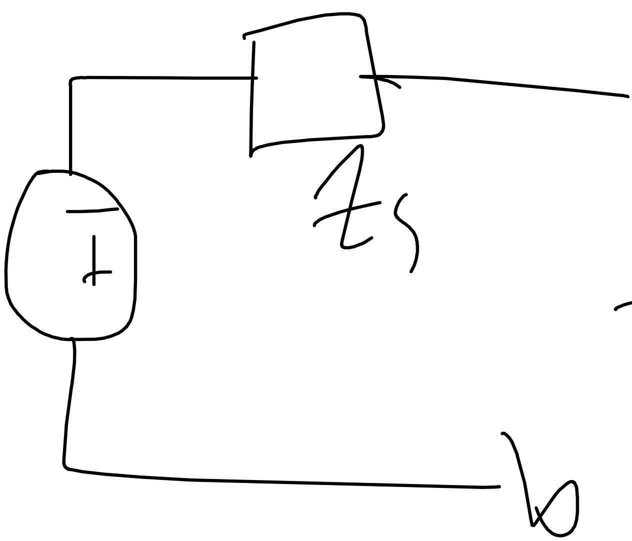
i_{o1}

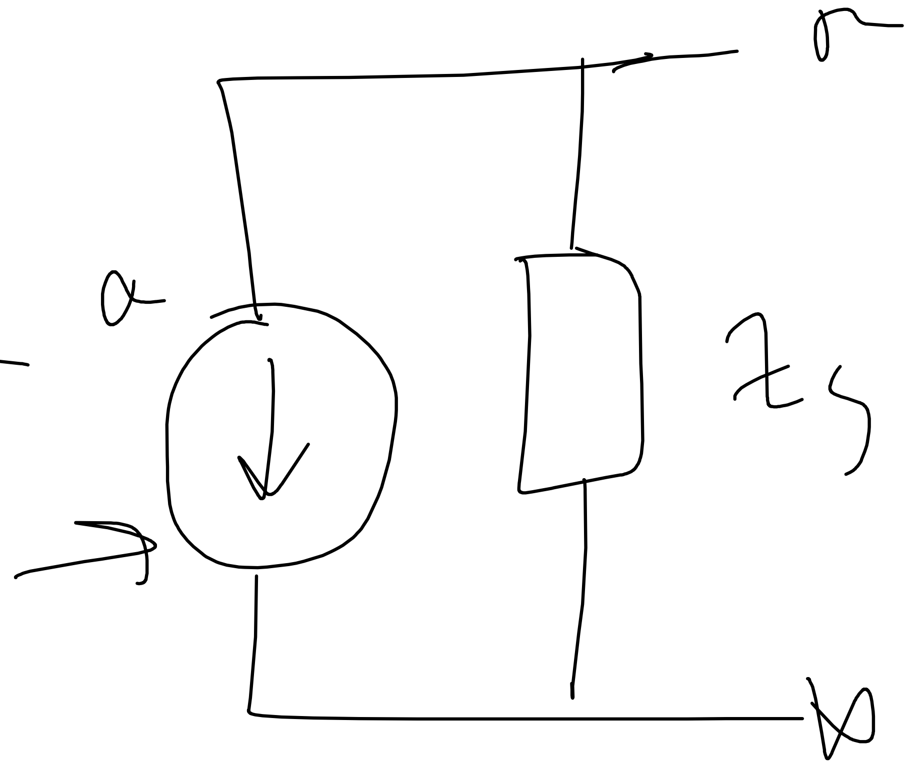
i_{o2}

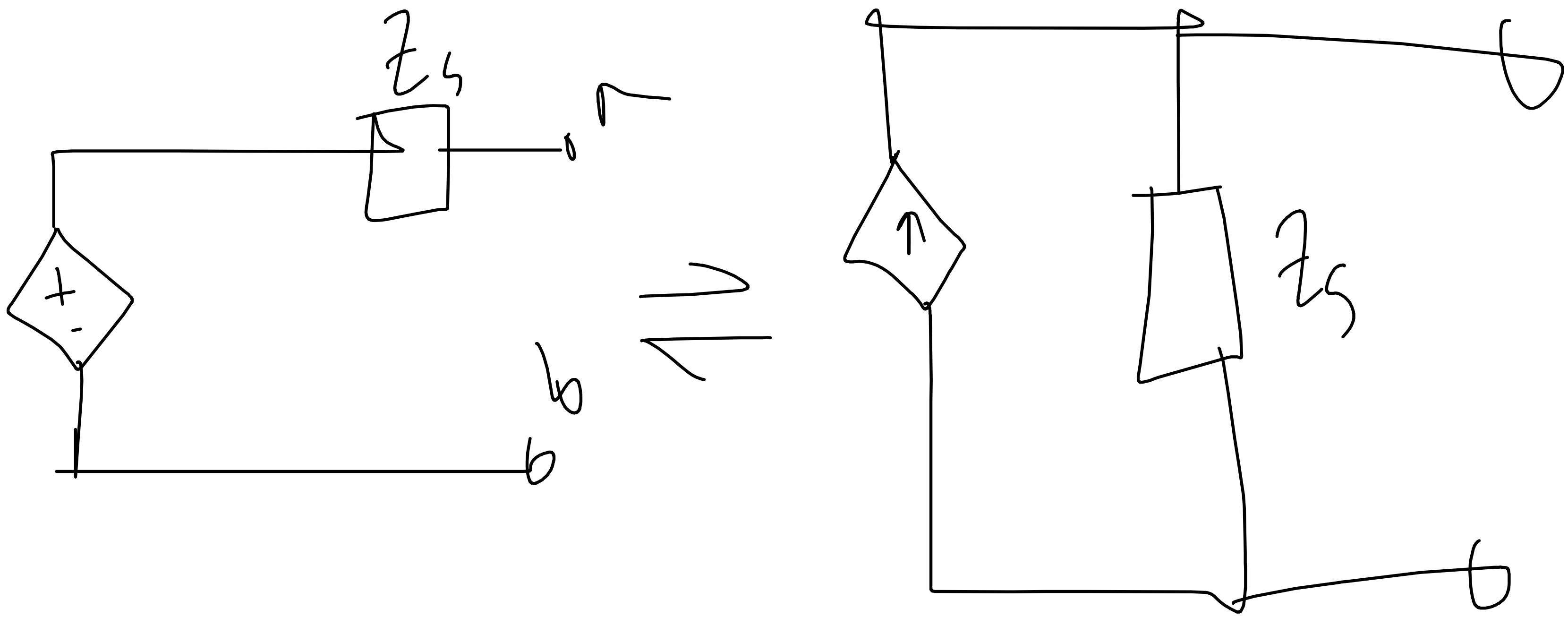
In supperposition you $\boxed{\text{cannot}}$ turn off the $\boxed{\text{dependent source}}$

Source Transformation



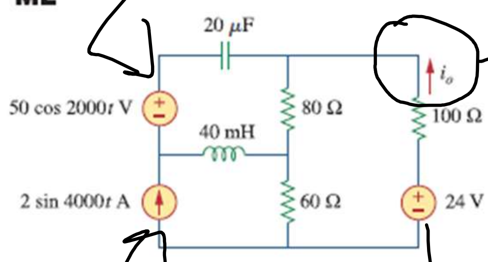





Zs
+
-
b
Zs

10.48 Find i_o in the circuit of Fig. 10.93 using superposition.

ML



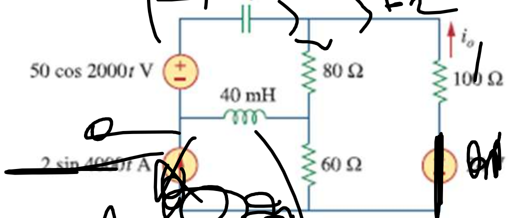
i_{o1}

i_{o2}

i_{o3}

10.48 Find i_o in the circuit of Fig. 10.93 using superposition.

ML



i_{o1}

$j80$

$$Z_T = \left(\left(100+60\right) \| 80\right) + \left(j80 - j25\right)$$

$$20\mu F \longrightarrow \frac{1}{wC}$$

$$= \frac{1}{2000 \times 20\mu F} = 25\,\Omega$$

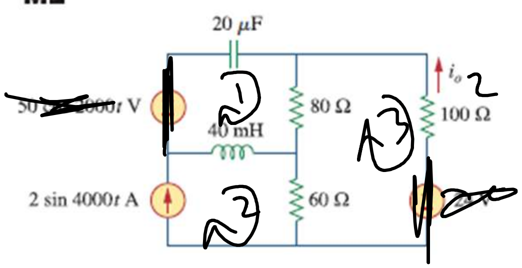
$$I_1 = \frac{50\angle 0^\circ}{Z_T}$$

$$I_2 = I_1 \times \frac{80}{160+80}$$

$$=$$

$$I_{o_{II}} = -I_2 \qquad \boxed{i_{o1}}$$

10.48 Find i_o in the circuit of Fig. 10.93 using superposition.



@ i_{o2}

Applying Mesh.

I_1

I_2

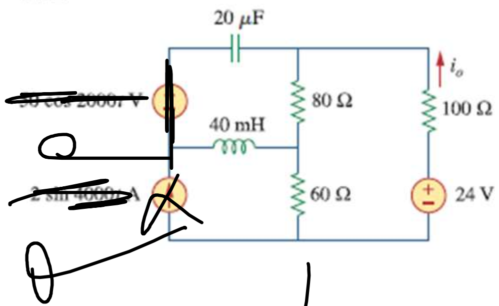
I_3

$I_{o2} = -I_3$

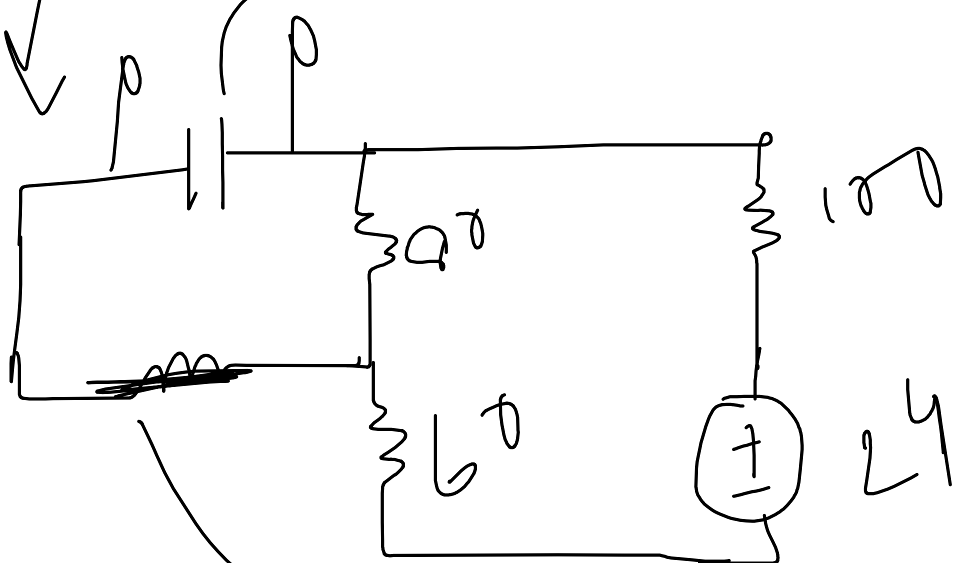
i_{o2}

10.48 Find i_o in the circuit of Fig. 10.93 using superposition.

ML



i_{o2}



$20\mu F \rightarrow \dfrac{1}{2\pi \cancel{f}^{\,0} \times 20\mu} = \infty$

$40\ mH \rightarrow 2\pi \cancel{f}^{\,0} \times 40m \rightarrow 0$

$$i_{o3} = \frac{24}{240} \quad 0.1A$$

$$= 0.1\,A$$

$$i_o = i_{o1} + i_{o2} + i_{o3}$$

$$=$$